

Priyadharshini Manikandan

PL/SQL Developer/BI Analyst

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Location : New Jersey, USA

PL/SQL and Power BI Engineer with 8 years of experience in database development with expertise in business application analysis, design, and implementation, good in PL/SQL, Data Analytics, Visualization (Power BI, Tableau).

- I have 5 years of experience in PL/SQL development, constructing physical and logical relational data models and interpreting requirements, designing and requirements documents for PL/SQL
- I have 3 years of experience in Power BI Development and extensive working experience in Data warehousing and Business Intelligence Technologies with expertise in Oracle, SQL, and Power BI for building, deploying, and managing applications.
- I have 2 years of experience in Data Analysis, Business Analysis, and Quality Assurance in IT
- Proficiency in SQL performance tuning using SGA and PGA optimization techniques to enhance database efficiency.
- Hands-on experience with JSON format handling in PL/SQL, integrating unstructured data into structured formats for analytics and reporting.
- Write, maintain, and uphold coding standards for SQL and PL/SQL stored program units and packages.
- Proficiency in query optimization, performance tuning, and debugging to enhance performance.
- Create unit tests (both code and documentation) for the stored program units in PL/SQL.
- Considerable expertise and competence in all facets of database programming on Oracle SQL, PL/SQL, and associated technologies that encompass SQL, such as collections, partitioned tables, triggers, procedures, functions, and packages, Oracle supplied packages, cursors, refer-cursors, and table indexing.
- Proficiency with Oracle distributed query performance tuning using EXPLAIN_PLAN and AUTOTRACE.
- Proficiency in utilizing bulking strategies, analytical functions and Oracle PL/SQL collections.
- Experience in creating objects such as Stored Procedures, Views, Triggers, User-defined functions, cursors, derived tables, and common table expressions (CTEs).
- Extensive knowledge of tuning PL/SQL queries to improve database performance.
- Experience developing end-to-end Power BI reporting solutions for large clients including data modeling, dashboards, SQL queries and ETL processing, workflow, and transformations.
- Excellent understanding of Data Warehousing methodologies and concepts, including Star schema, ETL processes, Dimensional modeling, and reporting tools.
- Build solution-driven dashboards by developing different chart types including Crosstab's, Heat/Geo/Tree Maps, Pie/Bar Charts, Line/Area Charts, Scatter Plots, Bullet Graphs, and Histograms in Power BI Pro. Deployed the Power BI reports to Power BI services and pinned visuals to make dashboards out of it.
- Good experience in creating, reviewing, and managing test cases.
- Excellent analytical skills to understand Business Requirements and to translate them to System Requirement Specifications.
- Applied statistical methods and hypothesis testing for trend analysis.
- Used python library like Matplotlib and Seaborn for advanced data visualization. Built machine learning models for predictive analytics and forecasting.
- Good in Agile Methodology
- Hands-on experience in ML model creation, Gen AI solutions and RAG

EDUCATION

- Masters in Science, Stream - Medical Bioinformatics | Sri Ramachandra Medical College, Chennai, India | 2013 - 2015
- Bachelors in Science, Stream - Bio-Technology | Prince Sri Venkateshwara College, Chennai, India | 2010 - 2013

SKILLS

- **Programming:** Python (Pandas, NumPy, Scikit-learn), Django, Flask, FastAPI
- **Databases & Query Optimization:** Oracle, PL/SQL, MySQL, Stored Procedures, Indexing, Performance Tuning, Data Migration
- **Data Analytics & Business Intelligence:** Power BI
- **Data Engineering:** Feature Engineering, Dimensionality Reduction, Data Preprocessing
- **ETL & Data Engineering:** SQL-based ETL, Data Preprocessing, and Feature Engineering.
- **Web Development:** Fast API, Django, Flask
- **AI/ML Frameworks:** TensorFlow, Scikit-learn, XGBoost

(DAX, Data Modeling), Tableau, Excel, Data Storytelling, Statistical Analysis

- **Automation & Process Optimization:** Power Automate, CI/CD Pipelines, Workflow Automation
- **Data Analysis & Visualization:** Power BI(DAX), Tableau, Matplotlib, Seaborn, Power Automate, Excel, PowerPoint.
- **Cloud Platforms:** AWS SageMaker, AWS Bedrock, GCP

- **LLMs:** RAG, Agentic AI, LangChain, LangGraph, llamaIndex, Haystack, Prompt Engineering
- **NLP:** Sentiment Analysis, Topic Modeling, Text Classification, Chatbot, Summarization, Q&A
- **MLOps:** Docker, Kubernetes, CI/CD Pipelines, Model Deployment
- **Vector Databases:** ChromaDB, FAISS, Haystack, Pinecone
- **Performance Metrics:** ROC-AUC, Precision-Recall, Cross-Validation
- **Collaboration:** Cross-Functional Teams, Stakeholder Communication

WORK EXPERIENCE

Novalink Solutions, New Jersey, USA – PL/SQL Developer/BI Analyst

12/2022 – Present

Client: Fiserv Corp

- Designed and optimized PL/SQL stored procedures, functions, and indexes, improving query performance by 40% through SGA and PGA tuning strategies.
 - Created and managed tables, views, materialized views, and database triggers to ensure data integrity and audit compliance.
 - Developed complex SQL queries, correlated subqueries, and dynamic SQL for business applications.
 - Performed database performance tuning using hints like MATERIALIZE, PARALLEL, and NONPARALLEL.
 - Developed and maintained Python scripts for data processing using NumPy & Pandas.
 - Coordinated with front-end teams, providing optimized stored procedures and insights into data structure.
 - Extensively used advanced PL/SQL features such as Records, Tables, Object Types and Dynamic SQL & Ref Cursors.
 - Implemented robust error handling using Exception Handling for debugging and application stability.
 - Developed server-side PL/SQL scripts for data manipulation, validation, and materialized views for remote instances.
 - Automated SQL-based ETL workflows, integrating Oracle and SQL Server databases with Power BI for real-time analytics.
 - Designed and developed interactive Power BI dashboards to track financial trends, employee performance, and customer retention.
 - Created data models, calculated measures, and optimized DAX queries for enhanced dashboard performance.
 - Built Power BI reports using bar charts, pie charts, donut charts, gauge, waterfall charts, maps, filters, and slicers.
 - Combined data from SQL Server, Excel, CSV, and text files using Power Query for seamless data blending.
 - Performed advanced data transformations in Power Query, including merging, appending, pivoting, unpivoting and conditional columns.
 - Executed dashboard setups and data manipulation using DAX functions such as table functions, aggregations, and iteration functions.
 - Published Power BI reports to Power BI Services (Pro) with scheduled subscriptions and various sharing options.
- Tools:** Oracle 19c, Oracle 12c, Oracle SQL Developer, Toad, MySQL Workbench, Power BI, PyCharm, Jupyter.

BitMind Solutions, India – PL/SQL Developer/BI Analyst

07/2018 – 08/2022

Client: South Asian Banks through D&B

- Designed and managed relational databases using PL/SQL, creating tables, views, indexes, stored procedures, and triggers.
- Developed stored procedures, functions, triggers, and packages to support business applications.
- Created and managed materialized views and synonyms with invoker's privileges to optimize data access.
- Ensured data integrity and audit compliance through structured database design.
- Developed and optimized complex SQL queries, correlated subqueries, and dynamic SQL for backend testing, data validation, and report generation.
- Improved query execution time by 15% through indexing, query optimization, and performance tuning techniques.
- Used Bulk Collections and FORALL to enhance query performance and reduce context switching between SQL and PL/SQL engines.
- Tuned PL/SQL packages and SQL queries using EXPLAIN PLAN, SQLTRACE, and optimizer hints for execution.
- Implemented exception handling for robust error management, debugging, and application stability.
- Designed, developed, and validated ETL processes for data extraction, transformation, and loading (ETL).

- Used SQL*Loader to load data from text files and spreadsheets into Oracle databases.
- Created and optimized indexes on tables for faster data retrieval, enhancing database performance.
- Extensively used optimizer hints to direct execution plans for optimal performance.
- Tuned PL/SQL packages and SQL queries, analyzing execution plans and creating indexes as needed.
- Provided database support for financial and operational reporting, ensuring data integrity.
- Developed and debugged PL/SQL code, creating, executing, and optimizing SQL queries.
- Supported production defect resolution and daily business transaction issues.
- Extensively used advanced PL/SQL features such as Records, Collections, Object Types, and Dynamic SQL.
- Designed and developed interactive Power BI dashboards and reports tailored to business needs.
- Created data models in Power BI, cleaning, transforming, and merging data from multiple sources (SQL Server, MS Access, Excel, CSV, and text files).
- Built data visualizations using DAX functions, including table functions, aggregations, and iteration functions.
- Utilized Power Query for advanced data transformation, including merging, appending, unpivoting, and conditional columns.
- Developed reports using bar charts, pie charts, donut charts, gauge charts, waterfall charts, maps, filters, and slicers.
- Migrated and optimized existing BIR reports into Power BI based on user requests.
- Conducted data analysis to uncover business insights, improving decision-making efficiency.
- Visualized client data using Tableau, Matplotlib, and Seaborn for better business understanding.
- Assisted in developing machine learning models for predictive analytics and AI-driven business intelligence.
- Processed BIR (Business Information Report) requests for South Asian clients, collaborating with D&B sister concerns.
- Regularly interacted with user groups to discuss requirements, updates, and performance improvements.
- Collaborated with stakeholders to refine data visualization strategies and improve reporting processes.
- Maintained daily MIS reports, trackers, and revenue reports for the regional sales manager.
- Trained and mentored a team of five analysts, enhancing SQL querying, Power BI usage, and data visualization skills.
- Provided guidance on data analytics best practices, leading to a 10% increase in team productivity.
- Onboarded and trained new team members, enabling them to become active contributors.
- Managed cross-functional collaboration with 57 bankers across the South Asian region.

Tools: Oracle 12c, Oracle 11g, Oracle SQL Developer, Toad, MySQL Workbench, Power BI, Tableau, Excel, Jupyter.

Dun & Bradstreet, India – Data Analyst

06/2017 – 06/2018

Client: South Asian Banks

- Collaborated with multiple teams across the processes and worked in the South Asia Client service team where we managed 57 bankers across the South Asian regime.
- Key role is to Process the client's request on BIR report (Product of D&B) Where we coordinate with various D&B sister concerns across the globe and deliver the reports.
- Research and investigation of data. Handling data from different file formats like CSV, Excel, JSON, Parquet and MySQL tables
- Analyze different data formats using Python and Excel.
- Visualization of data collected using Tableau, Matplotlib, and Seaborn in Python for better insight into Clients.
- Performed data validations and data references in the database in MySQL.
- As a member of that team, I have been training the new peers and making them an active contributor to the team.
- Allowed to prepare daily MIS, tracker, and revenue report to the regional sales manager.

Tools: MySQL, MySQL Workbench, Power BI, Excel.

CERTIFICATION:

- Base SAS 9 | AWS Machine Learning Associate | Data Analytics using ML & AI | Python for Data Science

REFERENCES:

- **GitHub** - <https://github.com/PriyadharshiniManikanadan>
- **LinkedIn** - <https://www.linkedin.com/in/priyadharshini-manikandan/>